

Table 16: Full zero-shot learning results on ETT datasets. A lower value indicates better performance. **Red**: the best, **Blue**: the second best.

Methods		TIME-LLM		LLMTime		GPT4TS		DLinear		PatchTST		TimesNet		Autoformer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
<i>ETTh1</i> → <i>ETTh2</i>	96	0.279	0.337	0.510	0.576	0.335	0.374	0.347	0.400	0.304	0.350	0.358	0.387	0.469	0.486
	192	0.351	0.374	0.523	0.586	0.412	0.417	0.447	0.460	0.386	0.400	0.427	0.429	0.634	0.567
	336	0.388	0.415	0.640	0.637	0.441	0.444	0.515	0.505	0.414	0.428	0.449	0.451	0.655	0.588
	720	0.391	0.420	2.296	1.034	0.438	0.452	0.665	0.589	0.419	0.443	0.448	0.458	0.570	0.549
	Avg	0.353	0.387	0.992	0.708	0.406	0.422	0.493	0.488	0.380	0.405	0.421	0.431	0.582	0.548
<i>ETTh1</i> → <i>ETTh2</i>	96	0.189	0.293	0.646	0.563	0.236	0.315	0.255	0.357	0.215	0.304	0.239	0.313	0.352	0.432
	192	0.237	0.312	0.934	0.654	0.287	0.342	0.338	0.413	0.275	0.339	0.291	0.342	0.413	0.460
	336	0.291	0.365	1.157	0.728	0.341	0.374	0.425	0.465	0.334	0.373	0.342	0.371	0.465	0.489
	720	0.372	0.390	4.730	1.531	0.435	0.422	0.640	0.573	0.431	0.424	0.434	0.419	0.599	0.551
	Avg	0.273	0.340	1.867	0.869	0.325	0.363	0.415	0.452	0.314	0.360	0.327	0.361	0.457	0.483
<i>ETTh2</i> → <i>ETTh1</i>	96	0.450	0.452	1.130	0.777	0.732	0.577	0.689	0.555	0.485	0.465	0.848	0.601	0.693	0.569
	192	0.465	0.461	1.242	0.820	0.758	0.559	0.707	0.568	0.565	0.509	0.860	0.610	0.760	0.601
	336	0.501	0.482	1.328	0.864	0.759	0.578	0.710	0.577	0.581	0.515	0.867	0.626	0.781	0.619
	720	0.501	0.502	4.145	1.461	0.781	0.597	0.704	0.596	0.628	0.561	0.887	0.648	0.796	0.644
	Avg	0.479	0.474	1.961	0.981	0.757	0.578	0.703	0.574	0.565	0.513	0.865	0.621	0.757	0.608
<i>ETTh2</i> → <i>ETTh2</i>	96	0.174	0.276	0.646	0.563	0.253	0.329	0.240	0.336	0.226	0.309	0.248	0.324	0.263	0.352
	192	0.233	0.315	0.934	0.654	0.293	0.346	0.295	0.369	0.289	0.345	0.296	0.352	0.326	0.389
	336	0.291	0.337	1.157	0.728	0.347	0.376	0.345	0.397	0.348	0.379	0.353	0.383	0.387	0.426
	720	0.392	0.417	4.730	1.531	0.446	0.429	0.432	0.442	0.439	0.427	0.471	0.446	0.487	0.478
	Avg	0.272	0.341	1.867	0.869	0.335	0.370	0.328	0.386	0.325	0.365	0.342	0.376	0.366	0.411
<i>ETTh1</i> → <i>ETTh2</i>	96	0.321	0.369	0.510	0.576	0.353	0.392	0.365	0.415	0.354	0.385	0.377	0.407	0.435	0.470
	192	0.389	0.410	0.523	0.586	0.443	0.437	0.454	0.462	0.447	0.434	0.471	0.453	0.495	0.489
	336	0.408	0.433	0.640	0.637	0.469	0.461	0.496	0.494	0.481	0.463	0.472	0.484	0.470	0.472
	720	0.406	0.436	2.296	1.034	0.466	0.468	0.541	0.529	0.474	0.471	0.495	0.482	0.480	0.485
	Avg	0.381	0.412	0.992	0.708	0.433	0.439	0.464	0.475	0.439	0.438	0.457	0.454	0.470	0.479
<i>ETTh1</i> → <i>ETTh2</i>	96	0.169	0.257	0.646	0.563	0.217	0.294	0.221	0.314	0.195	0.271	0.222	0.295	0.385	0.457
	192	0.227	0.318	0.934	0.654	0.277	0.327	0.286	0.359	0.258	0.311	0.288	0.337	0.433	0.469
	336	0.290	0.338	1.157	0.728	0.331	0.360	0.357	0.406	0.317	0.348	0.341	0.367	0.476	0.477
	720	0.375	0.367	4.730	1.531	0.429	0.413	0.476	0.476	0.416	0.404	0.436	0.418	0.582	0.535
	Avg	0.268	0.320	1.867	0.869	0.313	0.348	0.335	0.389	0.296	0.334	0.322	0.354	0.469	0.484
<i>ETTh2</i> → <i>ETTh2</i>	96	0.298	0.356	0.510	0.576	0.360	0.401	0.333	0.391	0.327	0.367	0.360	0.401	0.353	0.393
	192	0.359	0.397	0.523	0.586	0.434	0.437	0.441	0.456	0.411	0.418	0.434	0.437	0.432	0.437
	336	0.367	0.412	0.640	0.637	0.460	0.459	0.505	0.503	0.439	0.447	0.460	0.459	0.452	0.459
	720	0.393	0.434	2.296	1.034	0.485	0.477	0.543	0.534	0.459	0.470	0.485	0.477	0.453	0.467
	Avg	0.354	0.400	0.992	0.708	0.435	0.443	0.455	0.471	0.409	0.425	0.435	0.443	0.423	0.439
<i>ETTh2</i> → <i>ETTh1</i>	96	0.359	0.397	1.179	0.781	0.747	0.558	0.570	0.490	0.491	0.437	0.747	0.558	0.735	0.576
	192	0.390	0.420	1.327	0.846	0.781	0.560	0.590	0.506	0.530	0.470	0.781	0.560	0.753	0.586
	336	0.421	0.445	1.478	0.902	0.778	0.578	0.706	0.567	0.565	0.497	0.778	0.578	0.750	0.593
	720	0.487	0.488	3.749	1.408	0.769	0.573	0.731	0.584	0.686	0.565	0.769	0.573	0.782	0.609
	Avg	0.414	0.438	1.933	0.984	0.769	0.567	0.649	0.537	0.568	0.492	0.769	0.567	0.755	0.591

Table 17: Full ablations on ETTh1 and ETTm1 in predicting 96 and 192 steps ahead (MSE reported).

Variant	Long-term Forecasting				Few-shot Forecasting			
	ETTh1-96	ETTh1-192	ETThm1-96	ETThm1-192	ETTh1-96	ETTh1-192	ETThm1-96	ETThm1-192
A.1 Llama (Default; 32)	0.362	0.398	0.272	0.310	0.448	0.484	0.346	0.373
A.2 Llama (8)	0.389	0.412	0.297	0.329	0.567	0.632	0.451	0.490
A.3 GPT-2 (12)	0.385	0.419	0.306	0.332	0.548	0.617	0.447	0.509
A.4 GPT-2 (6)	0.394	0.427	0.311	0.342	0.571	0.640	0.468	0.512
A.5 Llama (QLoRA; 32)	0.391	0.420	0.310	0.338	0.543	0.611	0.578	0.618
B.1 w/o Patch Reprogramming	0.410	0.412	0.310	0.342	0.498	0.570	0.445	0.487
B.2 w/o Prompt-as-Prefix	0.398	0.423	0.298	0.339	0.521	0.617	0.432	0.481
C.1 w/o Dataset Context	0.402	0.417	0.298	0.331	0.491	0.538	0.392	0.447
C.2 w/o Task Instruction	0.388	0.420	0.285	0.327	0.476	0.529	0.387	0.439
C.3 w/o Statistical Context	0.391	0.419	0.279	0.347	0.483	0.547	0.421	0.461

in both cases. TIME-LLM exhibits remarkable superiority in forecasting with limited data—a fact that becomes particularly salient when compared to GPT4TS.

Table 18: Efficiency comparison between model reprogramming and parameter-efficient fine-tuning (PEFT) with QLoRA (Dettmers et al., 2023) on ETTh1 dataset in forecasting two different steps ahead.

Length		ETTh1-96			ETTh1-336		
Metric		Trainable Param. (M)	Mem. (MiB)	Speed(s/iter)	Trainable Param. (M)	Mem. (MiB)	Speed(s/iter)
Llama (8)	QLoRA	12.60	14767	0.237	12.69	15982	0.335
	Reprogram	5.62	11370	0.184	5.71	13188	0.203
Llama (32)	QLoRA	50.29	45226	0.697	50.37	49374	0.732
	Reprogram	6.39	32136	0.517	6.48	37988	0.632

Table 19: Standard deviations of our approach and the second-best method (PatchTST) on all time series datasets for long-term forecasting.

Model	TIME-LLM		PatchTST (2023)	
Dataset	MSE	MAE	MSE	MAE
ETTh1	0.408 ± 0.011	0.423 ± 0.012	0.413 ± 0.001	0.430 ± 0.002
ETTh2	0.334 ± 0.005	0.383 ± 0.009	0.330 ± 0.002	0.379 ± 0.007
ETTm1	0.329 ± 0.006	0.372 ± 0.007	0.351 ± 0.006	0.380 ± 0.002
ETTm2	0.251 ± 0.002	0.313 ± 0.003	0.255 ± 0.003	0.315 ± 0.002
Weather	0.225 ± 0.009	0.257 ± 0.008	0.225 ± 0.001	0.264 ± 0.001
Electricity	0.158 ± 0.004	0.252 ± 0.007	0.161 ± 0.001	0.252 ± 0.001
Traffic	0.388 ± 0.001	0.264 ± 0.006	0.390 ± 0.003	0.263 ± 0.003
ILI	1.435 ± 0.011	0.801 ± 0.008	1.443 ± 0.012	0.797 ± 0.002

Table 20: Standard deviations of our TIME-LLM and the second-best method (N-HiTS) on M4 datasets for short-term forecasting.

Model	TIME-LLM			N-HiTS (2023a)		
Dataset	SMAPE	MAPE	OWA	SMAPE	MAPE	OWA
Yearly	13.419 ± 0.117	3.005 ± 0.011	0.789 ± 0.003	13.422 ± 0.009	3.056 ± 0.017	0.795 ± 0.010
Quarterly	10.110 ± 0.107	1.178 ± 0.009	0.889 ± 0.007	10.185 ± 0.107	1.180 ± 0.007	0.893 ± 0.001
Monthly	12.980 ± 0.102	0.963 ± 0.005	0.903 ± 0.001	13.059 ± 0.101	1.013 ± 0.007	0.929 ± 0.005
Others	4.795 ± 0.117	3.178 ± 0.012	1.006 ± 0.009	4.711 ± 0.117	3.054 ± 0.011	0.997 ± 0.012
Averaged	11.983 ± 0.011	1.595 ± 0.021	0.859 ± 0.002	12.035 ± 0.111	1.625 ± 0.012	0.869 ± 0.005

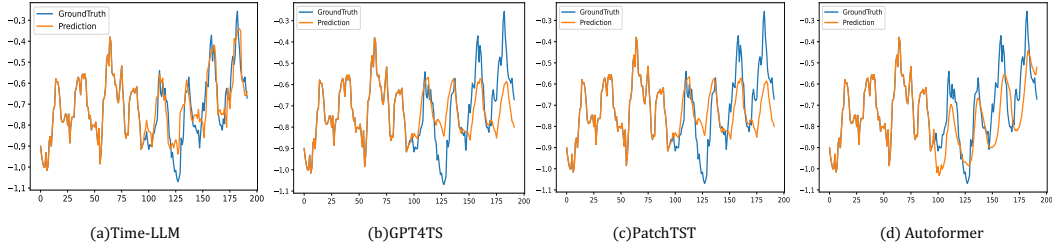


Figure 7: Long-term forecasting cases from ETTh1 by different models under the input-96-predict-96 settings. **Blue** lines are the ground truths and **orange** lines are the model predictions.

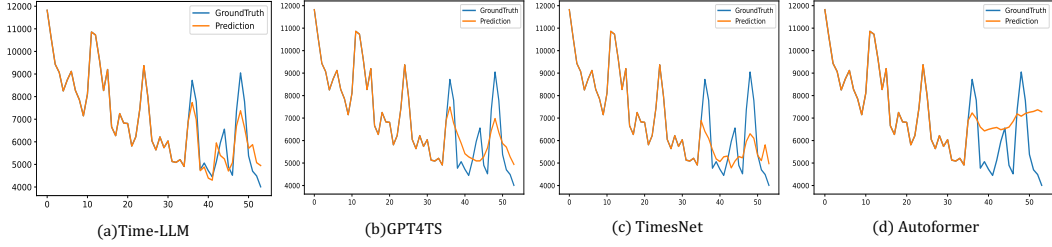


Figure 8: Short-term forecasting from the M4 dataset by different models under the input-36-predict-18 settings.

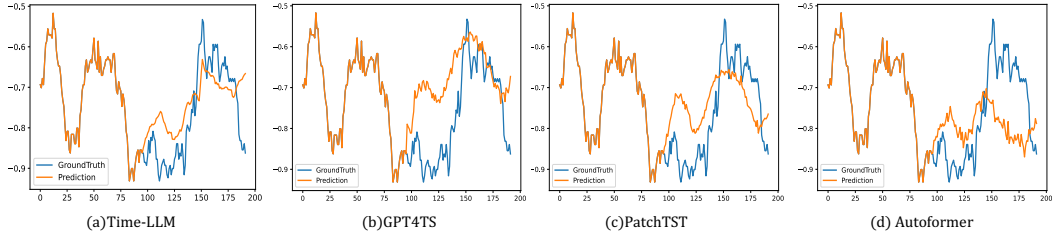


Figure 9: Few-shot forecasting cases from ETTm1 by different models under the input-96-predict-96 settings. **Blue** lines are the ground truths and **orange** lines are the model predictions.

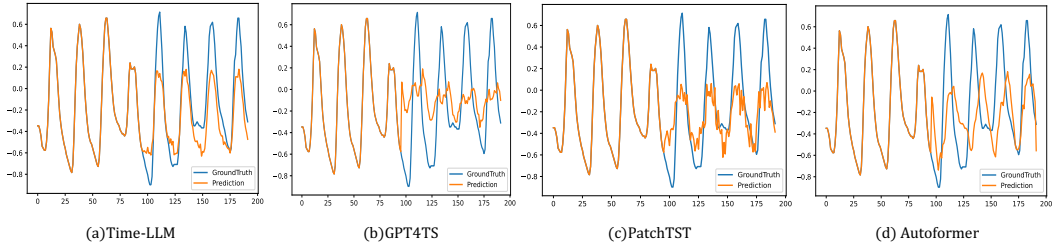


Figure 10: Zero-shot forecasting cases from ETTh1→ETTh2 by different models under the input-96-predict-96 settings. **Blue** lines are the ground truths and **orange** lines are the model predictions.